

FSLR - First Solar | Full Deep Dive

September 21, 2026 | Framework 6.0.2 | NOT HELD | TEMPORAL - domestic manufacturing and contract window | Watchlist

Analytical Base: 40/60 (67%). Valuation: 20x forward four-quarter GAAP-style operating profit.

Snapshot: USD 195.96 per share on September 18, 2026 close; 107.470 million shares, implying USD 21,059.78 million equity value. Source [FSLR-6]. Price is a dated header snapshot, not a valuation input in the body.

Active horizon Q3 2027 ceiling: USD 461.25/share. Q1 2028: USD 482.31/share.

All financial model amounts are USD millions, shares in millions, unless stated. Fiscal year ends December 31; quarter labels are calendar quarters.

1. Thesis, bottleneck and expanded TAM

First Solar supplies utility-scale CdTe modules. The relevant framework exposure is DB4 electricity supply, with a domestic-content contracting advantage. It is not an irreplaceable AI component: a developer can choose other generation or module technologies. The investable distinction is its qualified U.S. manufacturing footprint, its long-dated customer contracts and thin-film process IP. Classification is a temporal regulatory and backlog moat, rather than a permanent global module shortage.

Demand should be translated through contracted project watts and delivery dates. The Google-linked Steel River project is already in reported bookings; adding its watts again would double-count demand. AI electricity demand strengthens customer willingness to secure projects but does not establish a fixed percentage of hyperscaler capex for First Solar. Customer-level capex capture is unavailable, so this model uses the stronger company-specific watts/backlog bridge. [FSLR-3]

The key earnings distinction is between selling a module profitably before manufacturing incentives and recognizing domestic production credits. Q2's favorable tariff adjustment is also discrete. The model keeps credit economics visible and excludes another tariff-refund windfall. Domestic production is advantaged while qualifying rules persist; loss of those economics would require a materially lower operating path. [FSLR-1, FSLR-2]

The 2025 annual report's Competition section identifies intense crystalline-silicon competition, including low-price producers with state support. CdTe's process differentiation does not remove price-per-watt competition or guarantee technology leadership. TOPCon, bifacial silicon and eventual perovskite alternatives remain relevant. No additional small-cap competitor is added to this batch solely from this screen. [FSLR-5]

TAM expansion and S-curve

Phase	Economic exposure	Model treatment
Phase 1 - current, 100% relevance	Utility-scale domestic generation and contracted deliveries	Capacity- and backlog-constrained module revenue; no AI TAM multiplier
Phase 2 - 2026-2030, 50% conceptual weight	Additional load from data centers and electrification	New booking replenishment required; included in utilization assumptions only
Phase 3 - beyond 2027, 20% conceptual weight	Perovskite and adjacent next-generation photovoltaics	Pilot evidence exists; no standalone commercial sales added

S-curve: established manufacturing platform, with a 2026-2027 finishing-capacity step rather than a new zero-to-one market. The 2027 bridge is $20.1 \text{ GW} \times \$0.29/\text{W} = \$5,829\text{M}$ module revenue. A separate \$125M probability-weighted technology-adder pool takes total revenue to \$5,954M. Output remains below the combined physical fleet envelope; this does not mean every factory or credit regime is interchangeable. Beyond the current backlog, replacement bookings are an explicit requirement, not an assumed monopoly capture rate.

2. Latest evidence and financial anchor

Latest anchor: Q2 2026 and H1 results filed July 30. The fiscal guide is \$4.9-5.2B sales, \$2.4-2.6B gross profit and \$610-635M opex. Q3 module sales guidance is 3.9-4.5 GW. These ranges establish the near-term model floor and calibration. [FSLR-1]

Two-call management review: April 30 described India book-and-bill demand, CuRe pricing and constrained international utilization. July 30 retained the South Carolina first-phase start in H2 2026 but moved phase-two completion to mid-2027; the final finishing capability is 3.5 GW. The later call controls the ramp. Transcript spellings such as Q.O.R are treated as transcription artifacts, not separate products. [FSLR-3, FSLR-4]

Period	Status	Revenue	Gross profit	Operating profit	Source
Q1 2026	Actual, H1 less Q2	1,044.24	486.13	345.30	FSLR-1
Q2 2026	Actual	1,056.19	605.00	450.39	FSLR-1

The quarterly model reproduces FY2026 revenue of \$5,050M, gross profit of \$2,500M and opex of \$622.5M after including H1 actuals: all three equal guidance midpoints. H2 quarterly allocation is an analyst estimate, not additional guidance. Q3 volume is 4.2 GW, the center of the disclosed range. Latest actual revenue is copied from the filing rather than reconstructed from inferred geographic shares.

Q3 revenue is modeled at \$1,202.25M and Q4 at \$1,747.317M, the residual to full-year sales guidance. The model does not equate EBITDA with operating profit. Product margins in H2 are normalized to the full-year gross-profit midpoint after a separately disclosed manufacturing-benefit allocation; these margins are calibration assumptions because geographic product margins are not reported.

Balance sheet and capital allocation

At June 30 the company reported about \$1.7B net cash, against 2026 capex guidance of \$0.8-1.0B. That liquidity supports the manufacturing step, but cash consumption and customer advance-payment obligations still matter. Forecast shares increase modestly; no unannounced buyback is assumed. The ceiling convention does not add net cash separately. [FSLR-1, FSLR-2]

3. Base score and evidence ledger

Category	Score	Evidence
Core Thesis	16	Differentiated domestic platform; includes a two-point U.S. manufacturing consideration, with no global monopoly claim
Cycle Stage	7	South Carolina commercialization and 2027 throughput ramp
TAM Expansion	7	Power demand broadens customer opportunity; capacity constrains capture
Catalysts	4	H2 2026 start and mid-2027 phase-two milestone
Macro	8	Domestic supply-chain and load-growth demand, tempered by policy dependence
Alpha	2	Well-covered company; edge is separating credit and product economics
Risks	-4	Policy, execution, customer/project conversion, and competing technology
Total	40/60 = 67%	Watchlist

The score is 40/60, a Watchlist assessment. Domestic-content positioning earns credit, but neither backlog nor a subsidy establishes permanent scarcity. Four independent risk classes are deducted; the latest price does not affect Base.

4. Quarterly operating model

Single reported segment: modules. The model separates baseline module sales and the technology-adder pool, then shows production benefits outside sales. Gross profit is module revenue x underlying margin + technology-adder contribution + manufacturing benefits. R&D and SG&A are explicit revenue percentages; remaining startup opex is separately shown. Revenue is recognized at delivery, not at order signing.

Revenue vector	Driver and source	Confidence
Baseline module sales	Delivered GW x realized ASP. FY2026 guide calibrated; 2027 capacity step from FSLR-3, later utilization/ASP analyst extensions.	MEDIUM near term; LOW beyond 2027
Probability-weighted CuRe price adders	FSLR-2: up to \$500M potential; analyst 50% recognition over 2027-28, no duplicate volume.	LOW

Model assumptions, not management guidance: 2027-2030 module shipments are 20.1/21.2/21.8/22.0 GW; realized baseline ASP is \$0.290/\$0.298/\$0.305/\$0.305 per watt. Underlying module GM progresses 9%/10%/11%/12%, subject to commodity costs and factory yields. CuRe optionality uses 50% of the filing's up-to-\$500M potential across 2027-2028, \$125M each year, with 100% incremental GM as a price adder and no extra units. It is not added after 2028; later baseline ASP carries the technology mix. Uncontracted perovskite revenue is zero. [FSLR-2, FSLR-3]

Physical and commercial drivers

Year	Total delivered volume	Baseline ASP	Production-benefit GP (USD M)
2026 H2	Guidance residual	4.2 GW Q3	1,295.00 GP allocation
	2027 20.1 GW	\$0.290/W	2,419.00
	2028 21.2 GW	\$0.298/W	2,523.00
	2029 21.8 GW	\$0.305/W	2,591.00
	2030 22.0 GW	\$0.305/W	1,943.25

Vector revenue, gross margin and gross profit

For each row: vector gross profit = vector revenue x its own gross margin. The following tables expose the inputs; they are analyst model assumptions unless explicitly identified as reported or guided.

Baseline module sales

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	1,202.25	3.86%	46.41
Q4 2026	1,747.32	3.86%	67.45
Q1 2027	1,282.38	9.00%	115.41
Q2 2027	1,398.96	9.00%	125.91
Q3 2027	1,457.25	9.00%	131.15
Q4 2027	1,690.41	9.00%	152.14
Q1 2028	1,389.87	10.00%	138.99
Q2 2028	1,516.22	10.00%	151.62
Q3 2028	1,579.40	10.00%	157.94
Q4 2028	1,832.10	10.00%	183.21
Q1 2029	1,462.78	11.00%	160.91
Q2 2029	1,595.76	11.00%	175.53
Q3 2029	1,662.25	11.00%	182.85
Q4 2029	1,928.21	11.00%	212.10
Q1 2030	1,476.20	12.00%	177.14
Q2 2030	1,610.40	12.00%	193.25
Q3 2030	1,677.50	12.00%	201.30
Q4 2030	1,945.90	12.00%	233.51

Probability-weighted CuRe price adders

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	0.00	100.00%	0.00
Q4 2026	0.00	100.00%	0.00
Q1 2027	27.50	100.00%	27.50
Q2 2027	30.00	100.00%	30.00
Q3 2027	31.25	100.00%	31.25
Q4 2027	36.25	100.00%	36.25
Q1 2028	27.50	100.00%	27.50
Q2 2028	30.00	100.00%	30.00
Q3 2028	31.25	100.00%	31.25
Q4 2028	36.25	100.00%	36.25
Q1 2029	0.00	100.00%	0.00
Q2 2029	0.00	100.00%	0.00
Q3 2029	0.00	100.00%	0.00
Q4 2029	0.00	100.00%	0.00
Q1 2030	0.00	100.00%	0.00
Q2 2030	0.00	100.00%	0.00
Q3 2030	0.00	100.00%	0.00
Q4 2030	0.00	100.00%	0.00

Combined quarterly earnings

Quarter	Status	Revenue	Gross profit	R&D	SG&A	Other opex	OP	OP margin
Q3 2026	Model	1,202.25	646.41	72.13	50.49	10.68	513.10	42.68%
Q4 2026	Model	1,747.32	762.45	104.84	73.39	15.52	568.71	32.55%
Q1 2027	Model	1,309.88	675.09	85.14	61.56	3.00	525.39	40.11%
Q2 2027	Model	1,428.96	736.47	92.88	67.16	3.00	573.42	40.13%
Q3 2027	Model	1,488.50	767.15	96.75	69.96	3.00	597.44	40.14%
Q4 2027	Model	1,726.66	889.90	112.23	81.15	3.00	693.51	40.16%
Q1 2028	Model	1,417.37	721.55	85.04	63.78	0.00	572.72	40.41%
Q2 2028	Model	1,546.22	787.14	92.77	69.58	0.00	624.79	40.41%
Q3 2028	Model	1,610.65	819.94	96.64	72.48	0.00	650.82	40.41%
Q4 2028	Model	1,868.35	951.13	112.10	84.08	0.00	754.95	40.41%
Q1 2029	Model	1,462.78	730.93	87.77	65.83	0.00	577.33	39.47%
Q2 2029	Model	1,595.76	797.37	95.75	71.81	0.00	629.82	39.47%
Q3 2029	Model	1,662.25	830.60	99.73	74.80	0.00	656.06	39.47%
Q4 2029	Model	1,928.21	963.49	115.69	86.77	0.00	761.03	39.47%
Q1 2030	Model	1,476.20	604.66	88.57	66.43	0.00	449.66	30.46%
Q2 2030	Model	1,610.40	659.63	96.62	72.47	0.00	490.54	30.46%
Q3 2030	Model	1,677.50	687.11	100.65	75.49	0.00	510.97	30.46%
Q4 2030	Model	1,945.90	797.05	116.75	87.57	0.00	592.73	30.46%

Gross-profit adjustment outside product revenue

H2 2026 benefits are a \$1,295M calibrated production-benefit proxy, within the full-year gross-profit guide; other income-related grants mean it is not an exact isolated 45X disclosure. From 2027, fully integrated U.S. shipments receive a modeled \$0.17/W, South Carolina finishing \$0.07/W, and no credit is attached to other international deliveries. Credits reduce production costs and are not sales. The 2030 factor is 75%. These allocations require eligibility and realized output; no repeated \$88.6M tariff benefit is included. [FSLR-2]

Quarter	Additional GP
Q3 2026	600.00
Q4 2026	695.00
Q1 2027	532.18

Quarter	Additional GP
Q2 2027	580.56
Q3 2027	604.75
Q4 2027	701.51
Q1 2028	555.06
Q2 2028	605.52
Q3 2028	630.75
Q4 2028	731.67
Q1 2029	570.02
Q2 2029	621.84
Q3 2029	647.75
Q4 2029	751.39
Q1 2030	427.51
Q2 2030	466.38
Q3 2030	485.81
Q4 2030	563.54

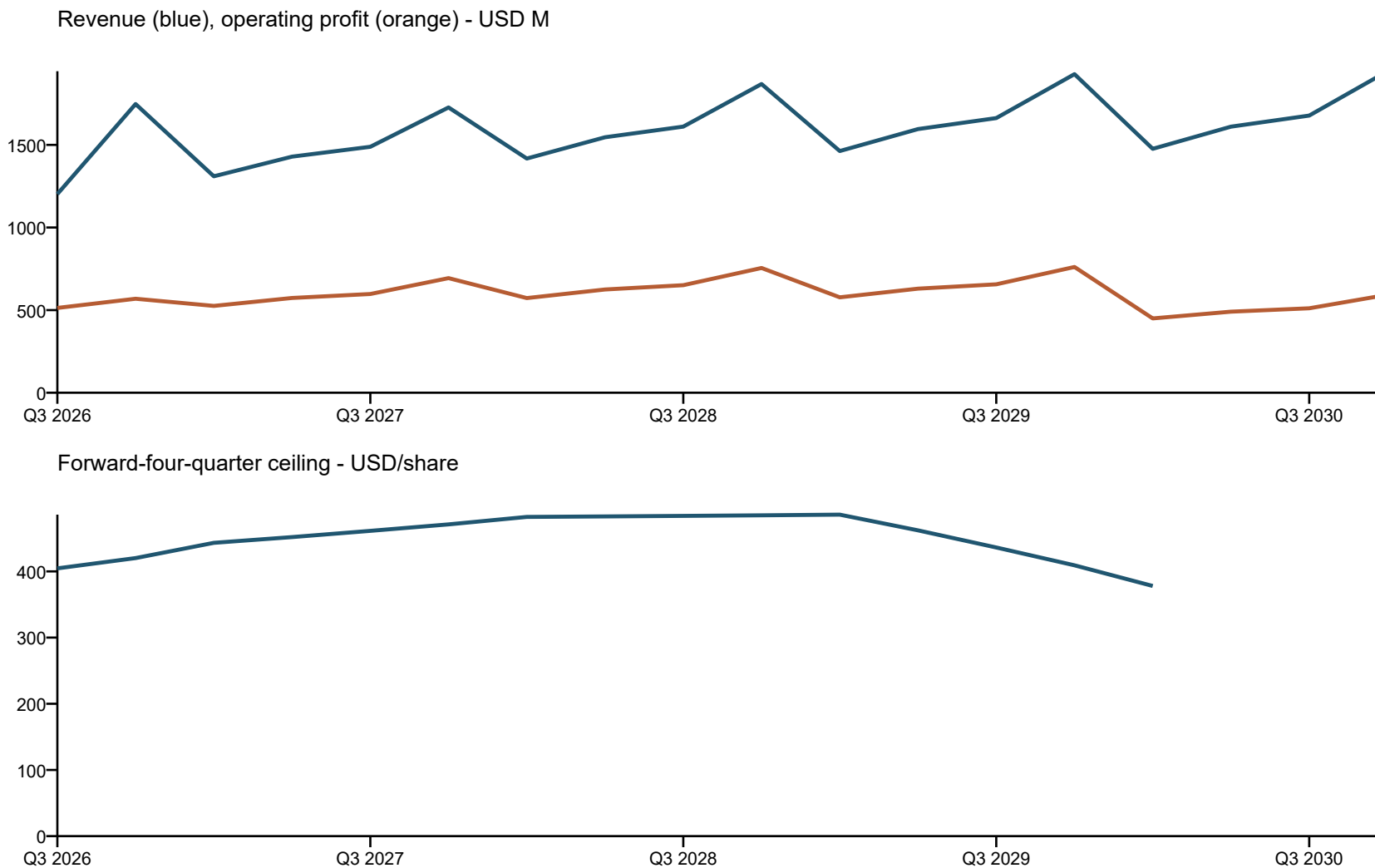
Annual trajectory cross-check

The calendar-bucket sums below are a capacity and demand cross-check; quarterly windows remain the live valuation method. Years beyond the latest guidance are model extensions, not company guidance.

Year	Revenue	Operating profit	OP margin
2027	5,954.00	2,389.76	40.14%
2028	6,442.60	2,603.29	40.41%
2029	6,649.00	2,624.24	39.47%
2030	6,710.00	2,043.90	30.46%

Sales assumptions approach the productive envelope without inventing another factory. The cumulative four-year shipments materially exceed today's backlog and therefore require new business. A higher future price or new capacity would need an updated contract or manufacturing source. The 2030 credit phase-down causes an explicit earnings decline even with stable shipments; the model does not smooth that inflection away.

5. Forward four-quarter ceiling and derivations



20x forward-four-quarter GAAP-style operating profit is the portfolio owner-selected multiple as of September 21, 2026, held constant across the model. This is the framework operating-profit convention, not an observed market P/E. A one-turn change changes each ceiling by 5%. Domestic-production credits remain in the existing GAAP-style operating model and retain their policy risk.

The framework convention applies the operating-profit multiple directly to diluted equity shares. It is neither a net-income P/E nor an enterprise-value bridge. No separate net-cash addition, tax conversion, probability haircut or price-dependent ranking enters this ceiling. Any probability-weighted revenue is already included once in its operating vector.

Ceiling quarter	Exact earnings window	Forward-4Q OP	Multiple	Diluted shares	Ceiling/share
Q3 2026	Q3 2026 to Q2 2027	2,180.62	20x	107.800	404.57
Q4 2026	Q4 2026 to Q3 2027	2,264.96	20x	107.825	420.12
Q1 2027	Q1 2027 to Q4 2027	2,389.76	20x	107.850	443.16
Q2 2027	Q2 2027 to Q1 2028	2,437.10	20x	107.875	451.84
Q3 2027	Q3 2027 to Q2 2028	2,488.46	20x	107.900	461.25
Q4 2027	Q4 2027 to Q3 2028	2,541.84	20x	107.925	471.04
Q1 2028	Q1 2028 to Q4 2028	2,603.29	20x	107.950	482.31
Q2 2028	Q2 2028 to Q1 2029	2,607.90	20x	107.975	483.06
Q3 2028	Q3 2028 to Q2 2029	2,612.93	20x	108.000	483.88
Q4 2028	Q4 2028 to Q3 2029	2,618.17	20x	108.025	484.73
Q1 2029	Q1 2029 to Q4 2029	2,624.24	20x	108.050	485.75
Q2 2029	Q2 2029 to Q1 2030	2,496.57	20x	108.075	462.01
Q3 2029	Q3 2029 to Q2 2030	2,357.29	20x	108.100	436.13
Q4 2029	Q4 2029 to Q3 2030	2,212.20	20x	108.125	409.19
Q1 2030	Q1 2030 to Q4 2030	2,043.90	20x	108.150	377.98

Complete derivation for every ceiling quarter

Q3 2026: OP window Q3 2026 to Q2 2027: $(513.10 + 568.71 + 525.39 + 573.42) \times 20 / 107.800 = \text{USD } 404.57$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2026: OP window Q4 2026 to Q3 2027: $(568.71 + 525.39 + 573.42 + 597.44) \times 20 / 107.825 = \text{USD } 420.12$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2027: OP window Q1 2027 to Q4 2027: $(525.39 + 573.42 + 597.44 + 693.51) \times 20 / 107.850 = \text{USD } 443.16$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2027: OP window Q2 2027 to Q1 2028: $(573.42 + 597.44 + 693.51 + 572.72) \times 20 / 107.875 = \text{USD } 451.84$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2027: OP window Q3 2027 to Q2 2028: $(597.44 + 693.51 + 572.72 + 624.79) \times 20 / 107.900 = \text{USD } 461.25$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2027: OP window Q4 2027 to Q3 2028: $(693.51 + 572.72 + 624.79 + 650.82) \times 20 / 107.925 = \text{USD } 471.04$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2028: OP window Q1 2028 to Q4 2028: $(572.72 + 624.79 + 650.82 + 754.95) \times 20 / 107.950 = \text{USD } 482.31$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2028: OP window Q2 2028 to Q1 2029: $(624.79 + 650.82 + 754.95 + 577.33) \times 20 / 107.975 = \text{USD } 483.06$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2028: OP window Q3 2028 to Q2 2029: $(650.82 + 754.95 + 577.33 + 629.82) \times 20 / 108.000 = \text{USD } 483.88$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2028: OP window Q4 2028 to Q3 2029: $(754.95 + 577.33 + 629.82 + 656.06) \times 20 / 108.025 = \text{USD } 484.73$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2029: OP window Q1 2029 to Q4 2029: $(577.33 + 629.82 + 656.06 + 761.03) \times 20 / 108.050 = \text{USD } 485.75$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2029: OP window Q2 2029 to Q1 2030: $(629.82 + 656.06 + 761.03 + 449.66) \times 20 / 108.075 = \text{USD } 462.01$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2029: OP window Q3 2029 to Q2 2030: $(656.06 + 761.03 + 449.66 + 490.54) \times 20 / 108.100 = \text{USD } 436.13$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2029: OP window Q4 2029 to Q3 2030: $(761.03 + 449.66 + 490.54 + 510.97) \times 20 / 108.125 = \text{USD } 409.19$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2030: OP window Q1 2030 to Q4 2030: $(449.66 + 490.54 + 510.97 + 592.73) \times 20 / 108.150 = \text{USD } 377.98$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

The final three operating quarters support earlier four-quarter windows and have no standalone ceiling. Arithmetic uses unrounded inputs; displayed components are rounded for readability.

Q2 weighted-average diluted shares were 107.732M. Model shares start at 107.8M and increase 0.025M per quarter for net compensation dilution. Outstanding shares in the header are a separate point-in-time measure. No split or buyback is forecast. [FSLR-1, FSLR-2]

Mechanical share sensitivity

Q1 2028 denominator	Ceiling/share
Current model	482.31
5% more shares	459.35
10% more shares	438.47

6. Cross-book relationships and monitoring

FSLR and SHLS share utility-project timing but sell different components; a common project is one end-demand pool. ENS can supply backup power, while grid equipment and transformer positions address a different part of the electricity constraint. Do not sum each supplier's revenue as a new market.

Paired watch: FSLR shipments and customer project schedules versus SHLS backlog conversion; domestic supply milestones versus continued need for grid interconnection and storage. An AI power forecast alone does not prove which supplier receives a contract.

7. Scored risks and re-rate tests

Risk class	Score	Failure channel
Policy and credits	-1	Loss of eligibility, tariff changes or weaker project incentives impair delivered economics.
Execution and costs	-1	Finishing delays, lower yield, freight or material inflation limit margins and volume.
Customer and project conversion	-1	Backlog rescheduling, terminations and permitting delays defer revenue; replenishment is necessary.
Competition and technology	-1	Silicon pricing or better alternatives reduce contracted and future pricing power.

Dated triggers and counters

Checkpoint	Positive evidence	Counter / model action
Q3 2026 report	Shipments within 3.9-4.5 GW and full-year range retained	Below-range output without a timing recovery cuts the quarterly path
H2 2026	South Carolina phase one produces saleable modules	Commissioning alone is insufficient; move recognition if shipment slips
Mid-2027	Phase two ready and utilization grows	Any delay pushes the physical shipment step to actual readiness
	2027-2028 CuRe adjusters enter recognized contract revenue	Reduce the 50% modeled realization if performance or notifications fail
Quarterly through 2030	Replacement bookings preserve delivered ASP and contract quality	Falling coverage or cancellations reduce utilization before reducing the multiple

The main uncertainty is the split between product margin and manufacturing benefits; reported geographic gross margins do not exist. This is a calibrated Plan-B model, with MEDIUM confidence through 2027 and LOW thereafter. No hypothetical policy outcome is treated as enacted law. The latest filings describe a 2030 phase-down, which is explicitly retained in the operating assumptions.

8. Sources and snapshot

- **FSLR-1** - 2026-07-30. Q2 2026 results and guidance, SEC exhibit. Actual financials, annual guidance, quarterly shipments and diluted shares.
- **FSLR-2** - 2026-07-30. Q2 2026 Form 10-Q. Manufacturing credits, revenue recognition, capital structure and technology-adder potential.
- **FSLR-3** - 2026-07-30. Q2 management earnings-call transcript. Management remarks only; capacity and contract timing. Third-party transcript of a primary event.
- **FSLR-4** - 2026-04-30. Q1 management earnings-call transcript. Management context on pricing, India, utilization and technology ramp.
- **FSLR-5** - 2025 annual report, filed 2026. First Solar 2025 Annual Report. Competition section and differentiated manufacturing.
- **FSLR-6** - 2026-09-18. Historical closing price. Dated market snapshot only.

Model snapshot: September 21, 2026. FSLR. Framework 6.0.2. Base 40/60 (67%). Eighteen modeled operating quarters and fifteen complete forward-four-quarter ceilings. Current full PDF and synchronized Markdown: G:/My Drive/1amInvesting/Artifacts/FSLR_FirstSolar_DeepDive.pdf and .md. The companion company worksheet uses this same model and links the ceiling path to the master portfolio.